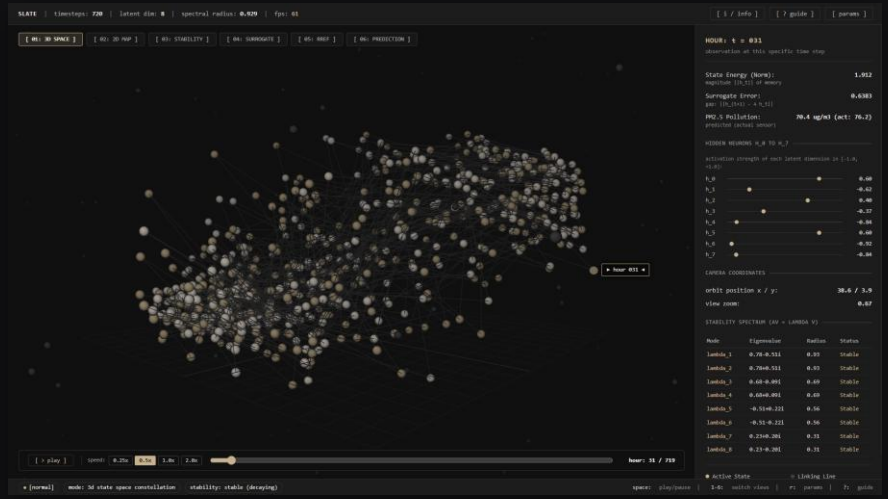
MATHS MINI PROJECT

SLATE

Spectral Latent Analysis & Trajectory Exploration

Opening the RNN black box with linear algebra and dynamical systems



Live SLATE dashboard: 720 hidden states in 3D

Presented by Thanas | Sudhanva | Sumanth | Sourabh

The RNN is a black box

Opaque memory

The recurrence hides how memory evolves hour by hour.

Blow-up or fade?

Without proof, activations may diverge or vanish over long runs.

Hidden redundancy

Do the 8 neurons store independent features, or copy each other?

SLATE's approach

Raw time series → Recurrent update → 8D latent state → Linear algebra analysis

Data and model setup

720 h
hourly steps, 30 full days

5
features, Z-score normalized

8
hidden neurons (latent dim)

THE RECURRENT STEP

$$h(t+1) = \tanh(W\,h(t) + U\,x(t) + b)$$

h(t) is in R^8. Stacking all hours gives the trajectory matrix H, size 720 x 8. Pure NumPy, W scaled to initial spectral radius 0.90.

INPUT FEATURES

- Carbon monoxide (CO)
- Nitrogen dioxide (NO2)
- Temperature
- Relative humidity
- Particulate matter (PM10)

Five linear algebra tools, one pipeline

1

QR basis

Orthonormal basis of the 8D state cloud

2

Orthogonal projection

Squash 8D down to 2D / 3D for viewing

3

Least squares

Fit a linear matrix A to the dynamics

4

Eigenvalues

Check |lambda| < 1 for stability

5

RREF

Test rank: any redundant neurons?

Each tool answers one question about the network, with a provable answer instead of a heuristic.

QR projection: seeing 8D in 3D

68-85%

of geometric variance kept in the 2D / 3D view

- Raw coordinates are correlated and cannot be inspected by eye.
- QR gives an exact orthonormal basis (Gram-Schmidt).
- P is symmetric and idempotent: a true orthogonal projector.

THE MATH

Decompose centered states

$$Hc' = Q\,R, \quad Q'Q = I$$

Keep first k columns (k = 2 or 3)

$$B = Q[:, :k]$$

Projector and coordinates

$$P = B\,B', \quad z(t) = B'(h(t) - mu)$$

Projector properties

$$P^2 = P, \quad P' = P$$

A linear surrogate captures 85% of the RNN

CAN ONE MATRIX A REPLACE THE TANH UPDATE?

Goal

```
h(t+1) ~ A h(t)
```

Overdetermined system (719 x 8)

```
H_next ~ H_prev A'
```

Least-squares solution (pseudoinverse)

```
A = (pinv(H_prev) H_next)'
```

Local neural memory behaves predominantly linearly.

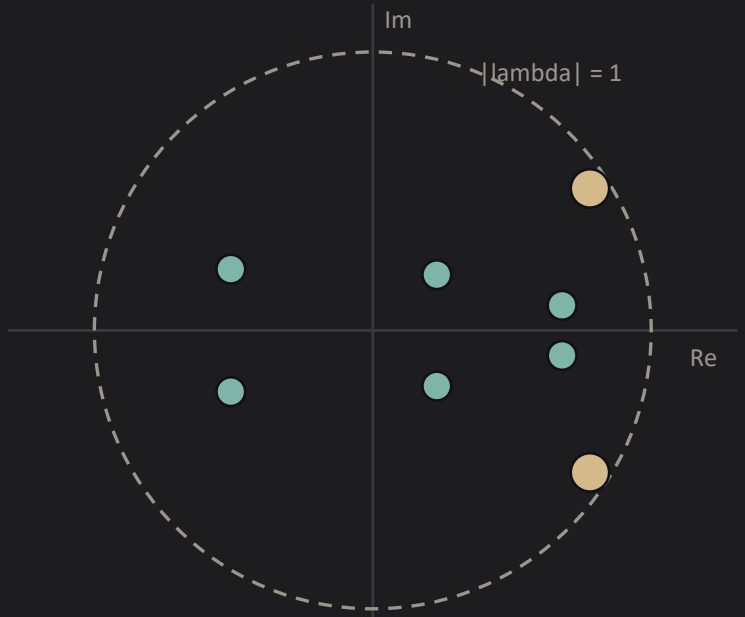
0.8495

R^2 of the linear fit (about 85%)

0.032

mean 1-step Euclidean residual

Provably stable: spectral radius < 1



Complex plane: 8 eigenvalues of A

0.9294

spectral radius, strictly less than 1

STABILITY CRITERION

`|lambda| < 1`	decays: stable sink
`|lambda| = 1`	neutral oscillation
`|lambda| > 1`	blows up: unstable

No redundancy: full rank 8 / 8

8 / 8

numerical rank (full rank)

METHOD

`rref(S), S is 8 x 8`

Gauss-Jordan elimination in SymPy on a state sample rounded to 4 decimals.

PIVOT COLUMNS

h0 pivot	**h1** pivot	**h2** pivot	**h3** pivot
h4 pivot	**h5** pivot	**h6** pivot	**h7** pivot

All 8 latent dimensions are pivots: zero free columns, so nothing can be pruned.

Downstream check: predicting PM2.5

0.9401

R^2 of the linear readout

3.355

RMSE in ug/m3

READOUT

$$y(t) = w' h(t) + b$$

The latent geometry encodes real daily pollution cycles

Morning rush

CO and NO2 build up; trajectory speeds up

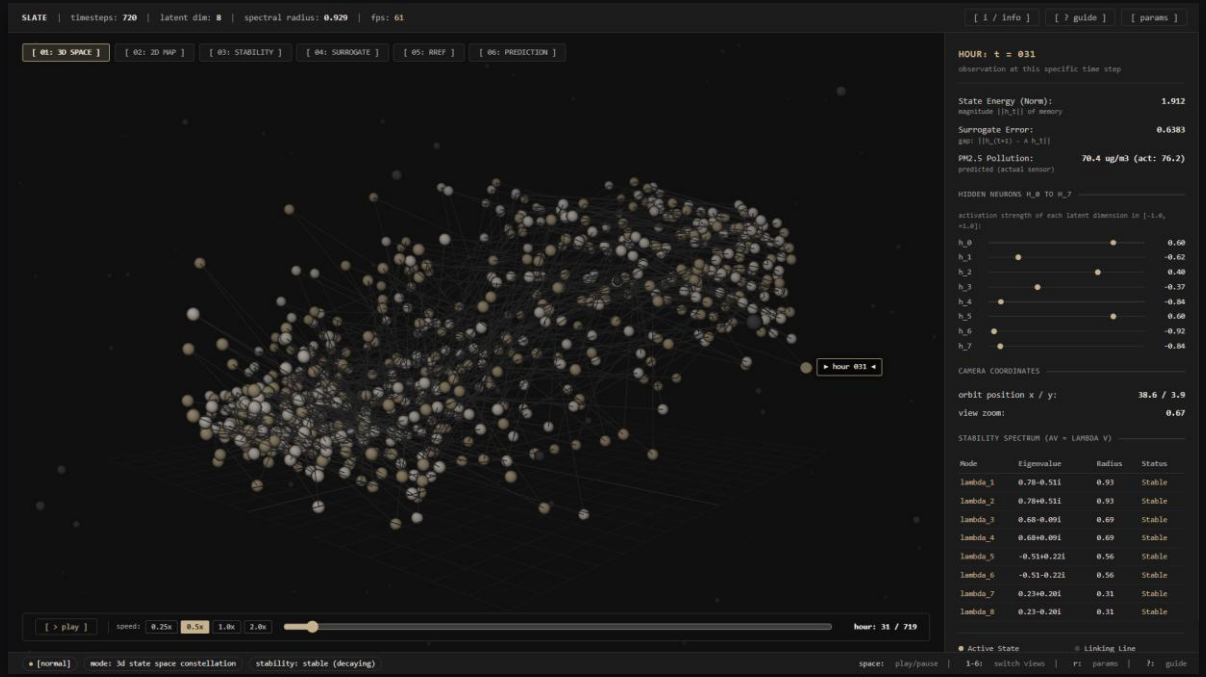
Afternoon spikes

Photochemical activity drives sharp peaks

Night baseline

States settle and contract toward the origin

Live demo: the SLATE dashboard



Python + FastAPI backend, Three.js frontend

01	**3D space** orbit, zoom, playback
02	**2D map** actual vs surrogate
03	**Stability** unit-circle spectrum
04	**Surrogate** hourly residual error
05	**RREF** pivots and full rank
06	**Prediction** sensor vs predicted PM2.5

Takeaways and what's next

WHAT WE SHOWED

Linear algebra, not guesswork
QR, pseudoinverse, eigenvalues, RREF

Linearizable
surrogate $R^2 = 0.8495$

Provably stable
spectral radius $0.9294 < 1$

No redundancy
rank 8/8; PM2.5 readout $R^2 = 0.94$

Beyond RNNs

Extend to Transformer residual streams and continuous-time Neural ODEs.

Smarter training

Track RREF rank online during training to detect and prune dead neurons.

Questions?

SLATE: Spectral Latent Analysis & Trajectory Exploration

```
Run locally:  python run.py   ->   http://127.0.0.1:8000
```